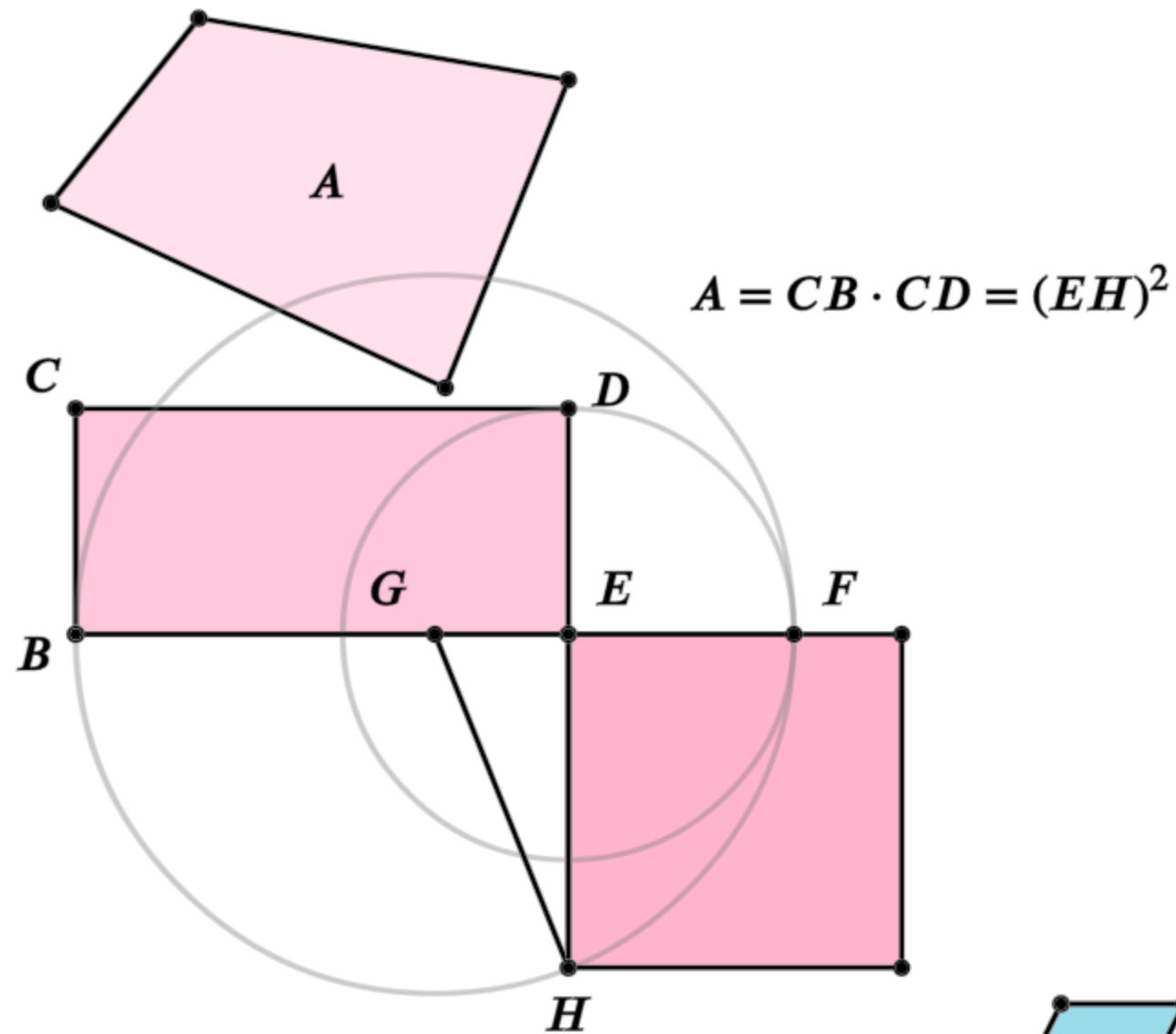


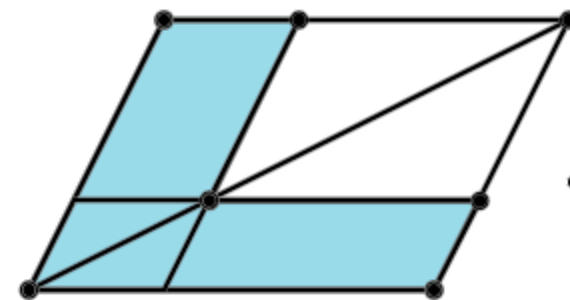
Euclid's Elements

Book II



It is a remarkable fact in the history of geometry, that the Elements of Euclid, written two thousand years ago, are still regarded by many as the best introduction to the mathematical sciences.

- Florian Cajori,
A History of Mathematics (1893)

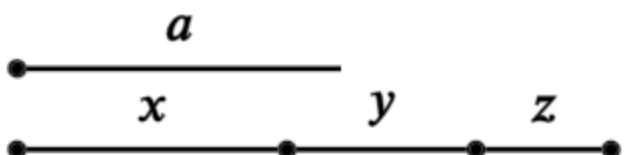


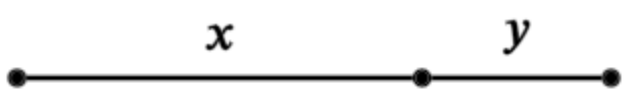
Definitions:

Any rectangular parallelogram is said to be contained by the two straight lines containing the right angle.

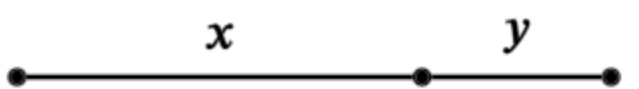
And in any parallelogrammic area let any one whatever of the parallelograms about its diameter with the two complements be called a gnomon.

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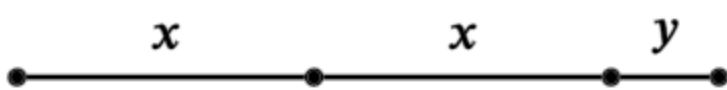
1. 
 $a(x + y + z) = ax + ay + az$

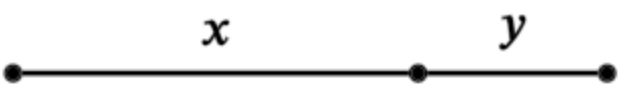
2. 
 $(x + y)^2 = x(x + y) + y(x + y)$

3. 
 $y(x + y) = yx + y^2$

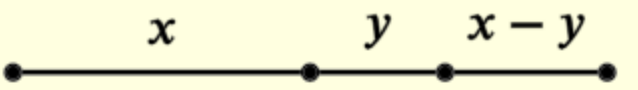
4. 
 $(x + y)^2 = x^2 + y^2 + 2xy$

5. 
 $(x + y)(x - y) + y^2 = x^2$

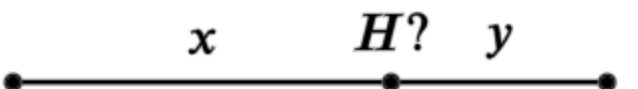
6. 
 $y(2x + y) + x^2 = (x + y)^2$

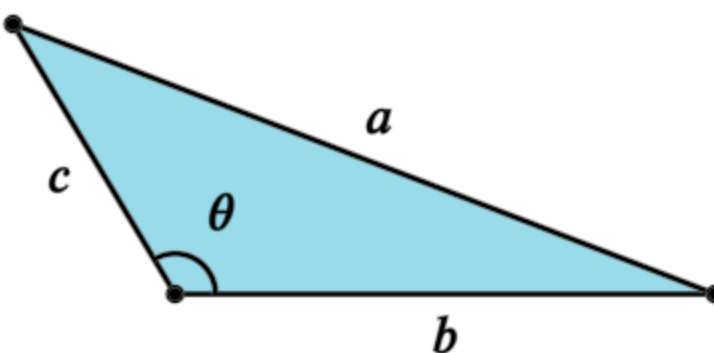
7. 
 $(x + y)^2 + y^2 = x^2 + 2y(x + y)$

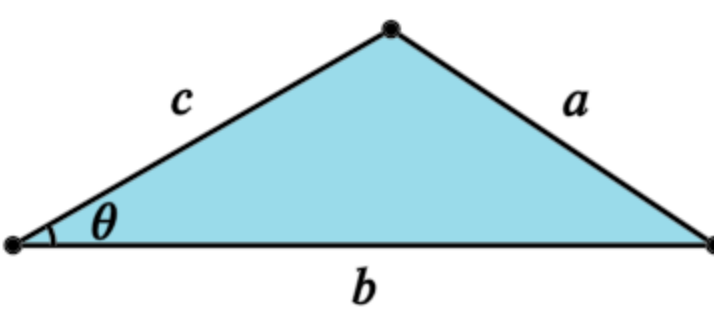
8. 
 $(x + 2y)^2 = 4y(x + y) + x^2$

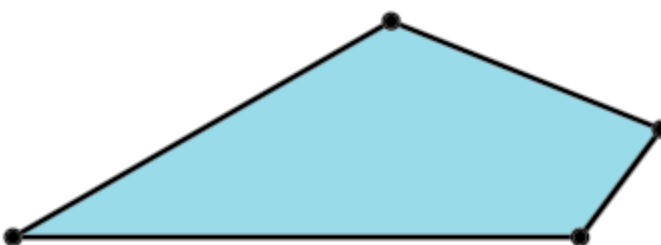
9. 
 $(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$

10. 
 $(2x + y)^2 + y^2 = 2(x^2 + (x + y)^2)$

11. 
Find H such that: $y(x + y) = x^2$

12. 
Cosine Law, $a^2 = b^2 + c^2 - 2bc \cos \theta$

13. 
Cosine Law. $a^2 = b^2 + c^2 - 2bc \cos \theta$

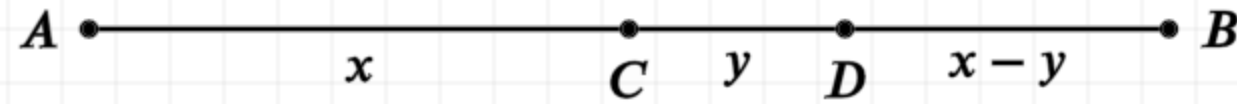
14. 
Construct a square equal to the polygon

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.

In other words

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D



$$AC = BC, \quad AB = AC + CD + DB$$

$$2x = x + y + (x - y)$$

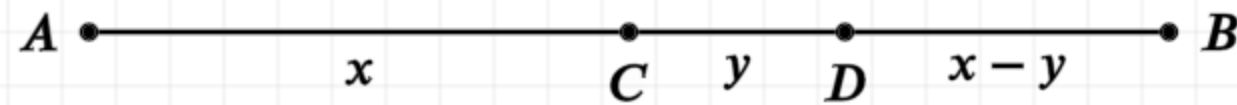
Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.

In other words

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

The sum of the squares of AD and DB is equal to twice the sum of the squares of AC and DC



$$AC = BC, \quad AB = AC + CD + DB$$

$$AD^2 + DB^2 = 2(AC^2 + CD^2)$$

$$2x = x + y + (x - y)$$

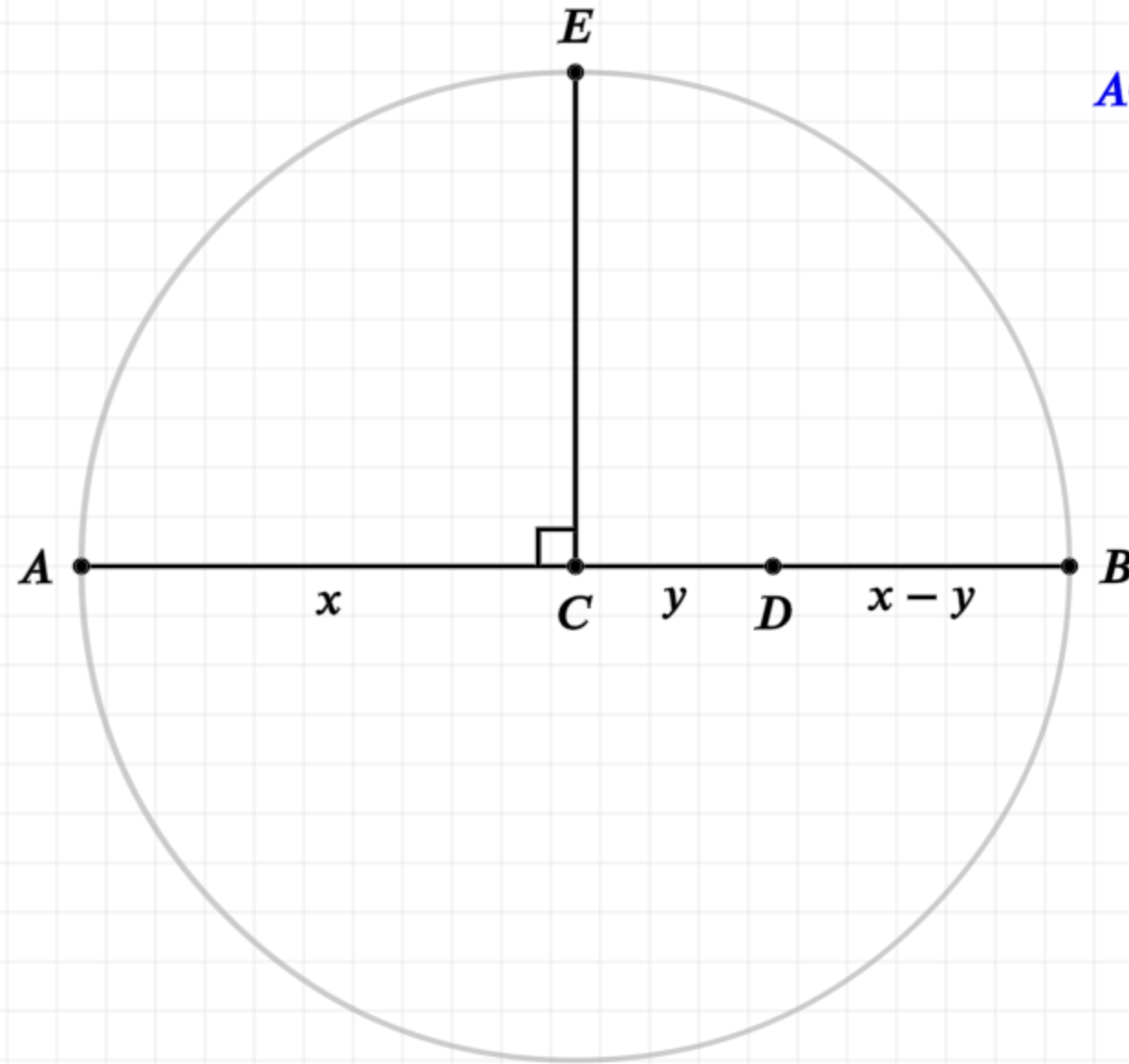
$$(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$$

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.

Construction:

Draw a line perpendicular to AB through point C (I.11), and make its length equal to AC {nb:or CB (I.3)}



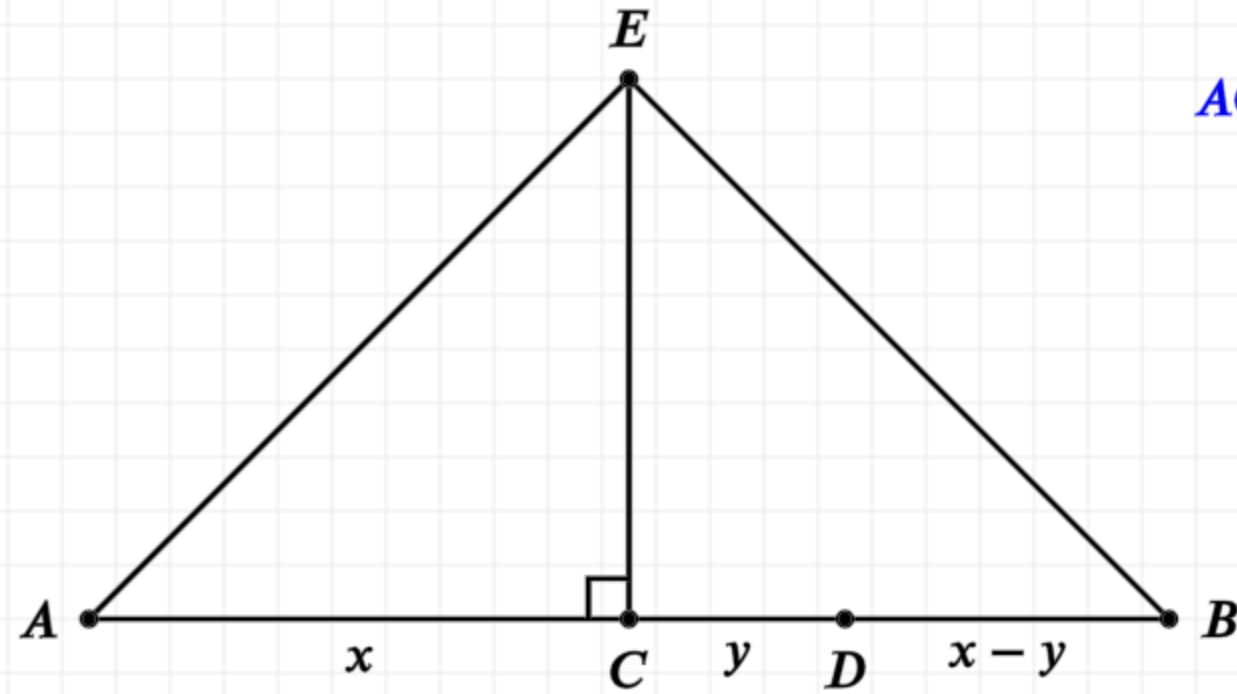
$$AC = BC, \quad AB = AC + CD + DB$$

$$AC = CE$$

$$CB = CE$$

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.



$$AC = BC, \quad AB = AC + CD + DB$$

$$AC = CE$$

$$CB = CE$$

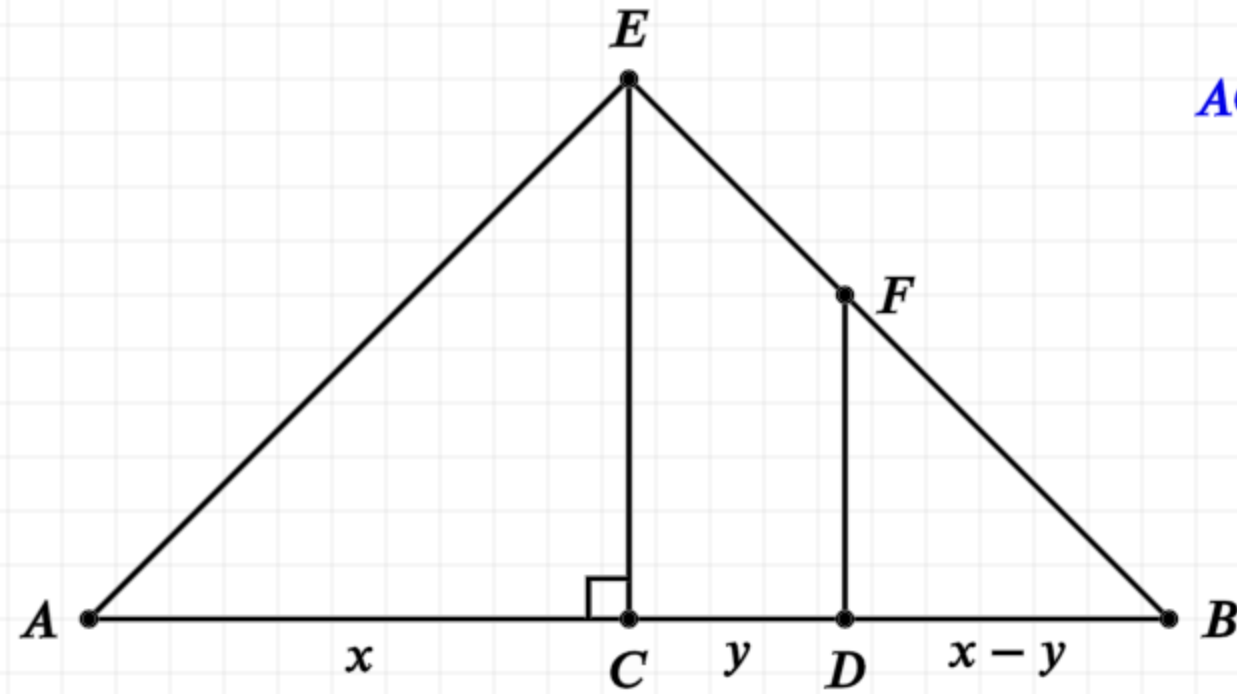
Construction:

Draw a line perpendicular to AB through point C (I.11), and make its length equal to AC {nb:or CB} (I.3)

Connect AE and EB

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.



$$AC = BC, \quad AB = AC + CD + DB$$

$$AC = CE$$

$$CB = CE$$

Construction:

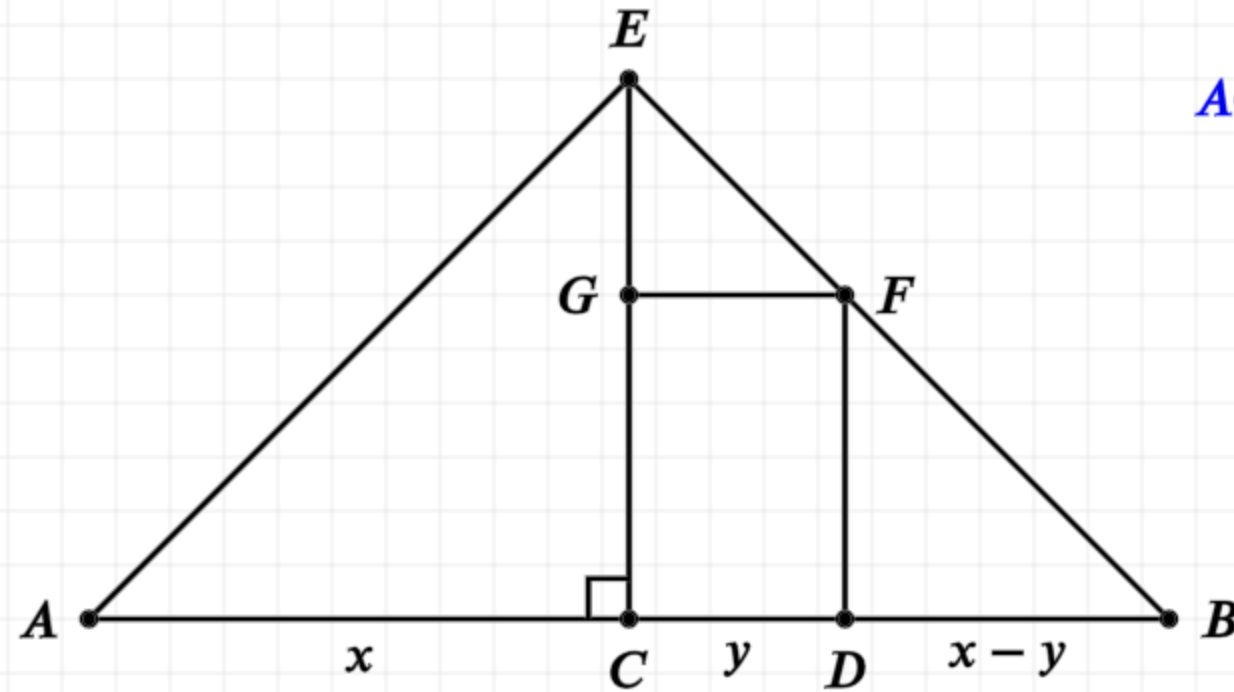
Draw a line perpendicular to AB through point C (I.11), and make its length equal to AC {nb:or CB } (I.3)

Connect AE and EB

Draw a line parallel to EC through point D

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.



$$AC = BC, \quad AB = AC + CD + DB$$

$$AC = CE$$

$$CB = CE$$

Construction:

Draw a line perpendicular to AB through point C (I.11), and make its length equal to AC {nb:or CB} (I.3)

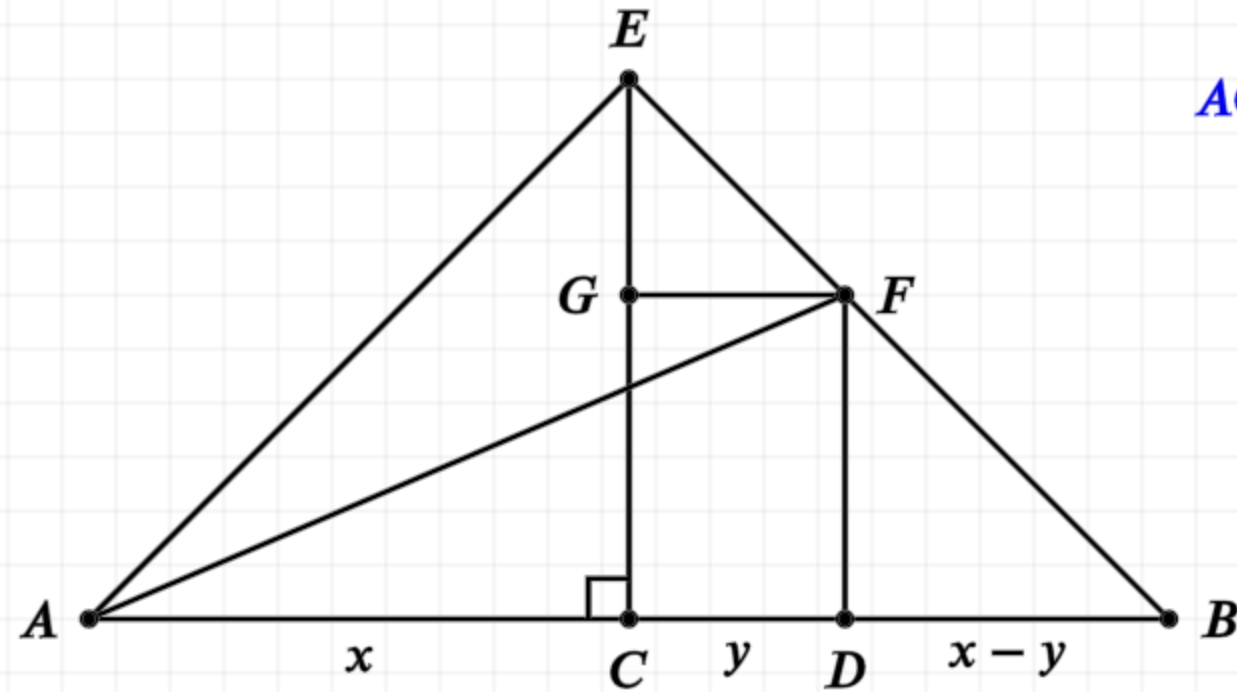
Connect AE and EB

Draw a line parallel to EC through point D

Draw a line parallel to AB through point F

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.



$$AC = BC, \quad AB = AC + CD + DB$$

$$AC = CE$$

$$CB = CE$$

Construction:

Draw a line perpendicular to AB through point C (I.11), and make its length equal to AC {nb:or CB} (I.3)

Connect AE and EB

Draw a line parallel to EC through point D

Draw a line parallel to AB through point F

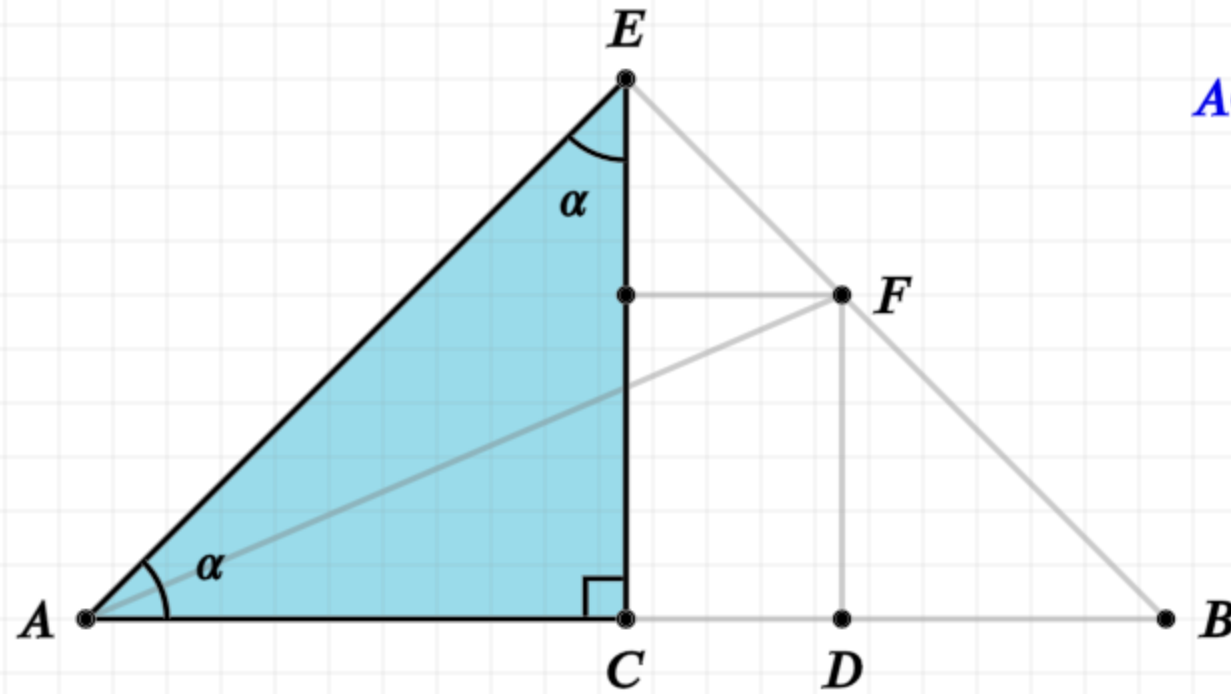
Join AF

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.

Proof (sides and angles)

Triangle AEC is a right angle triangle, and AC and CE are equal, therefore it is an isosceles triangle



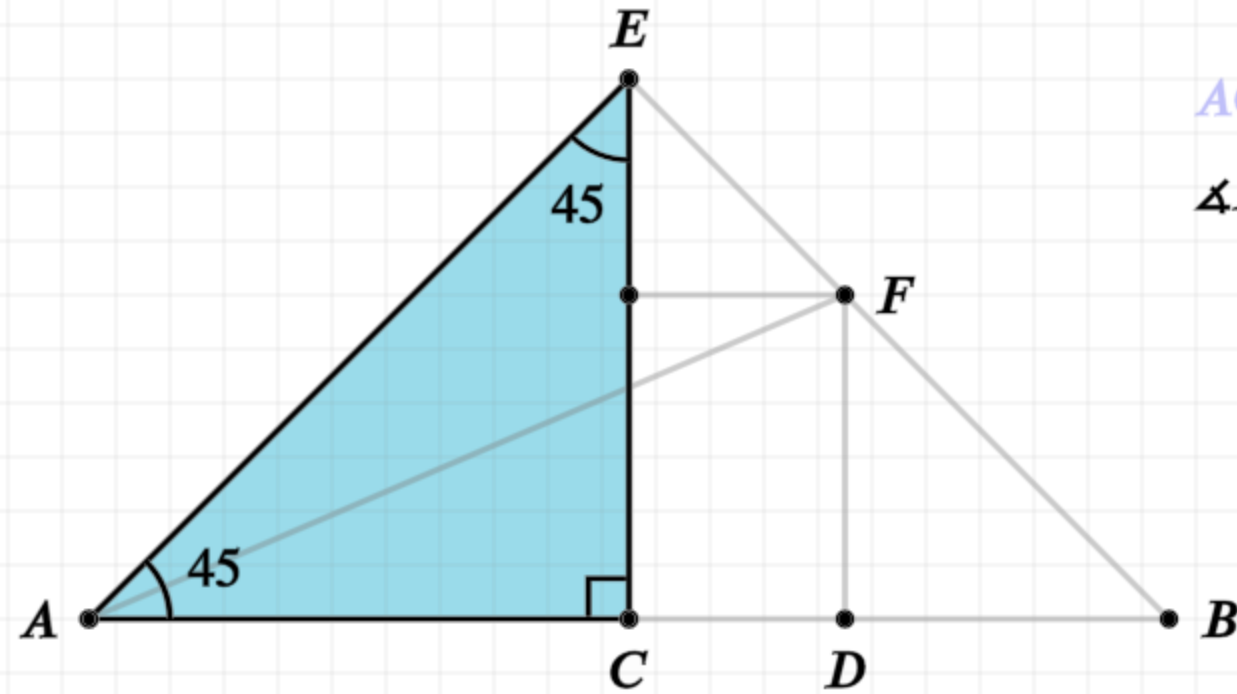
$$AC = BC, \quad AB = AC + CD + DB$$

$$AC = CE$$

$$CB = CE$$

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.



$$AC = BC, \quad AB = AC + CD + DB$$

$$\angle EAC = \angle CEA = 45^\circ$$

$$AC = CE$$

$$CB = CE$$

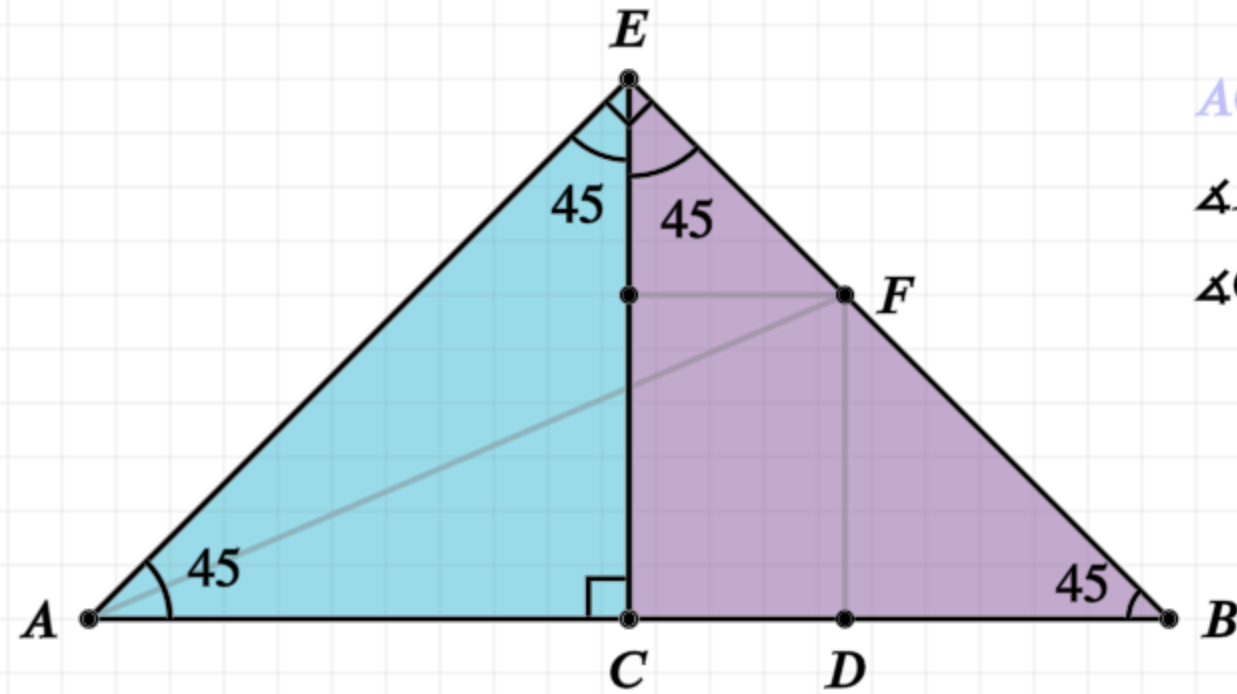
Proof (sides and angles)

Triangle AEC is a right angle triangle, and AC and CE are equal, therefore it is an isosceles triangle

Since the sum of the angles in a triangle equals two right angles (I.32), and ACE is a right angle, then the two base angles (being equal (I.5)) each equal one half a right angle (45 degrees)

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.



$$AC = BC, \quad AB = AC + CD + DB$$

$$\angle EAC = \angle CEA = 45^\circ$$

$$\angle CEB = \angle CBE = 45^\circ, \quad \angle AEB = \text{L}$$

$$AC = CE$$

$$CB = CE$$

Proof (sides and angles)

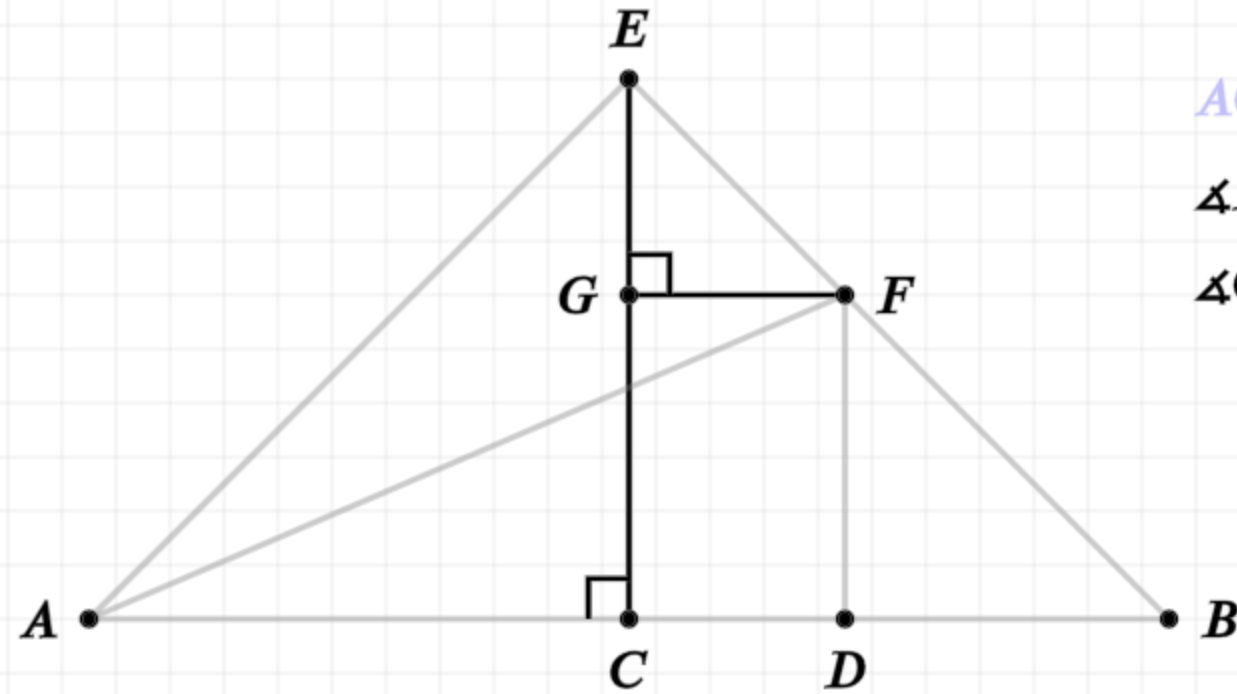
Triangle AEC is a right angle triangle, and AC and CE are equal, therefore it is an isosceles triangle

Since the sum of the angles in a triangle equals two right angles (I.32), and ACE is a right angle, then the two base angles (being equal (I.5)) each equal one half a right angle (45 degrees)

By the same reason, angles CEB and CBE are each half a right angle, which makes AEB a right angle

Proposition 9 of Book 2

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Proof (sides and angles)

Triangle AEC is a right angle triangle, and AC and CE are equal, therefore it is an isosceles triangle

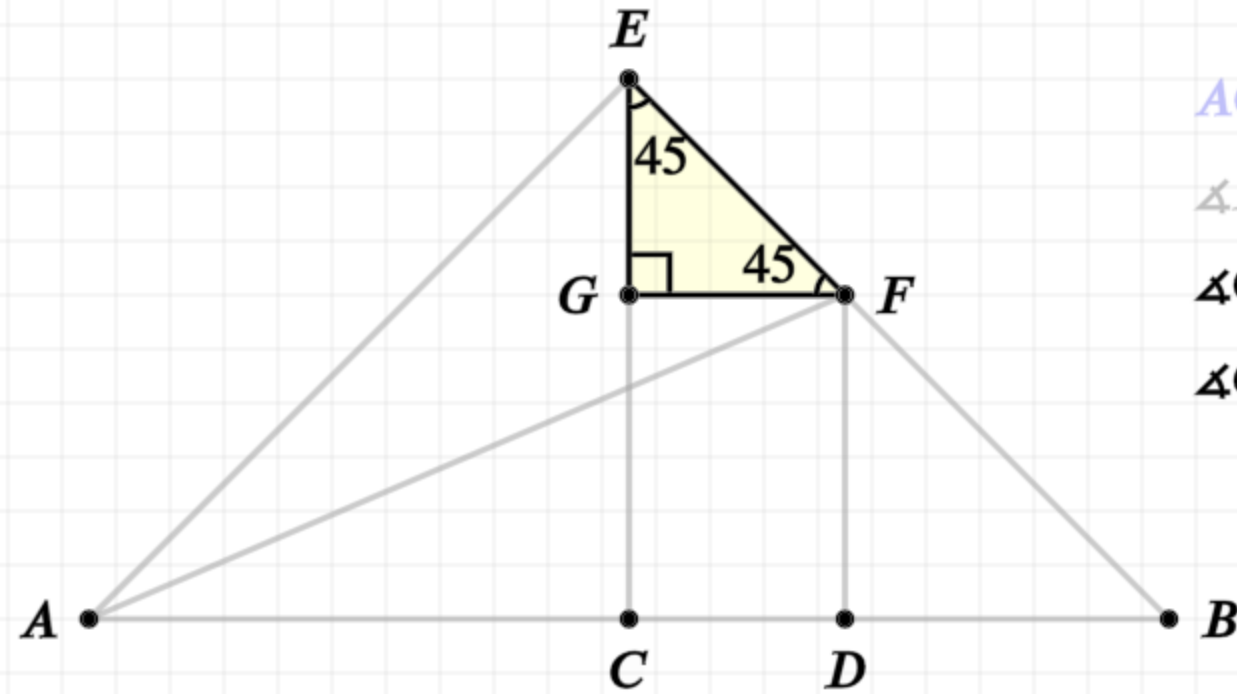
Since the sum of the angles in a triangle equals two right angles (I.32), and ACE is a right angle, then the two base angles (being equal (I.5)) each equal one half a right angle (45 degrees)

By the same reason, angles CEB and CBE are each half a right angle, which makes AEB a right angle

Since AB and GF are parallel, and CE intersects them, the opposite and interior angles are equal (I.29), so EGF is a right angle

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.



$$AC = BC, \quad AB = AC + CD + DB$$

$$\angle EAC = \angle CEA = 45^\circ$$

$$\angle CEB = \angle CBE = 45^\circ, \quad \angle AEB = 90^\circ$$

$$\angle GEF = \angle EFG = 45^\circ$$

$$AC = CE$$

$$CB = CE$$

$$EG = GF$$

Proof (sides and angles)

Triangle AEC is a right angle triangle, and AC and CE are equal, therefore it is an isosceles triangle

Since the sum of the angles in a triangle equals two right angles (I.32), and ACE is a right angle, then the two base angles (being equal (I.5)) each equal one half a right angle (45 degrees)

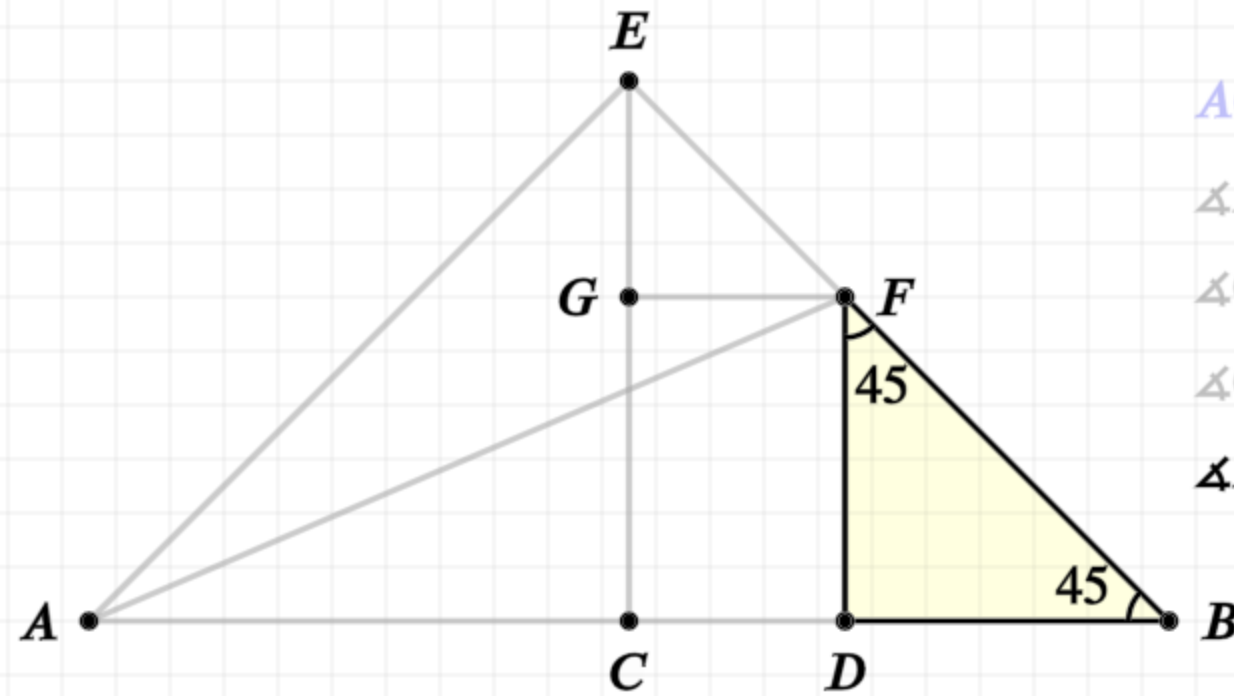
By the same reason, angles CEB and CBE are each half a right angle, which makes AEB a right angle

Since AB and GF are parallel, and CE intersects them, the opposite and interior angles are equal (I.29), so EGF is a right angle

The angle EFG is one half a right angle (I.32), and since two angles are equal, EGF is isosceles (I.6), so EG equals GF

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.



$$AC = BC, \quad AB = AC + CD + DB$$

$$\angle EAC = \angle CEA = 45^\circ$$

$$\angle CEB = \angle CBE = 45^\circ, \quad \angle AEB = 90^\circ$$

$$\angle GEF = \angle EFG = 45^\circ$$

$$\angle DFB = \angle FBD = 45^\circ$$

$$AC = CE$$

$$CB = CE$$

$$EG = GF$$

$$DB = FD$$

Proof (sides and angles)

Triangle AEC is a right angle triangle, and AC and CE are equal, therefore it is an isosceles triangle

Since the sum of the angles in a triangle equals two right angles (I.32), and ACE is a right angle, then the two base angles (being equal (I.5)) each equal one half a right angle (45 degrees)

By the same reason, angles CEB and CBE are each half a right angle, which makes AEB a right angle

Since AB and GF are parallel, and CE intersects them, the opposite and interior angles are equal (I.29), so EGF is a right angle

The angle EFG is one half a right angle (I.32), and since two angles are equal, EGF is isosceles (I.6), so EG equals GF

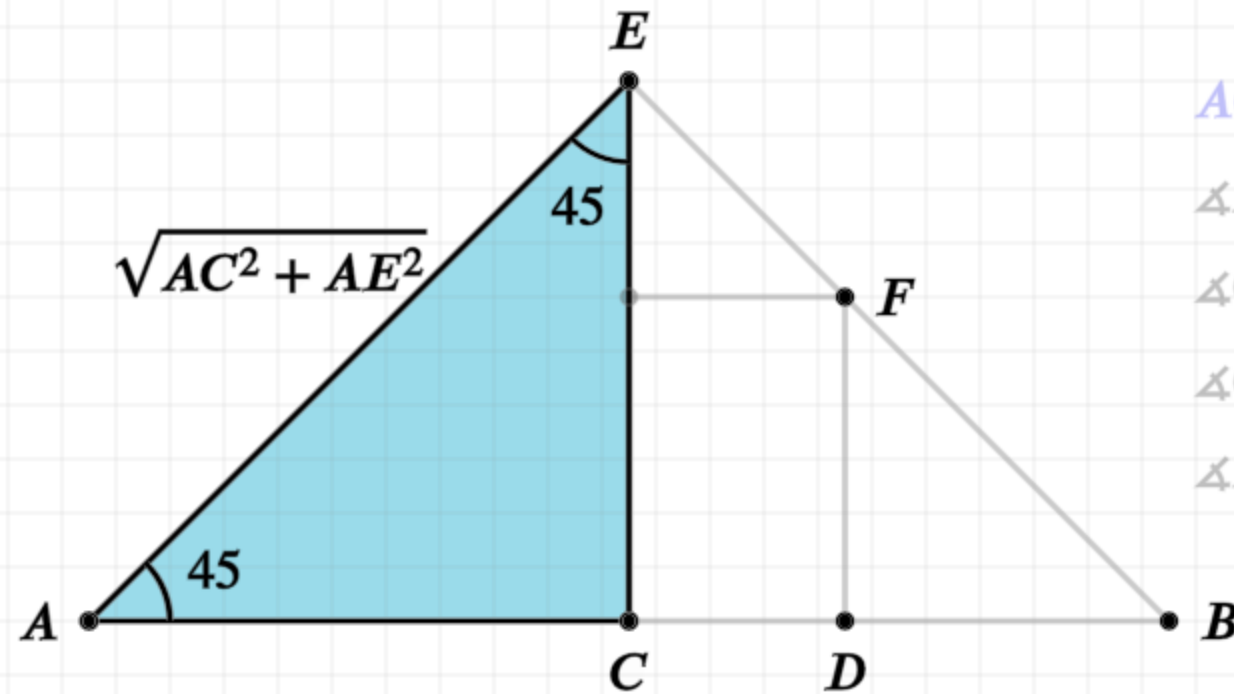
Using the same logic, FDB is also an isosceles triangle, and DB equals FD

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.

Proof (squares)

The triangle AEC is a right angle triangle, thus the square on AE equals the sum of the squares of AC and EC



$$AC = CB, \quad AB = AC + CD + DB$$

$$\angle EAC = \angle CEA = 45^\circ$$

$$\angle CEB = \angle CBE = 45^\circ, \quad \angle AEB = 90^\circ$$

$$\angle GEF = \angle EFG = 45^\circ$$

$$\angle DFB = \angle FBD = 45^\circ$$

$$AC = CE$$

$$CB = CE$$

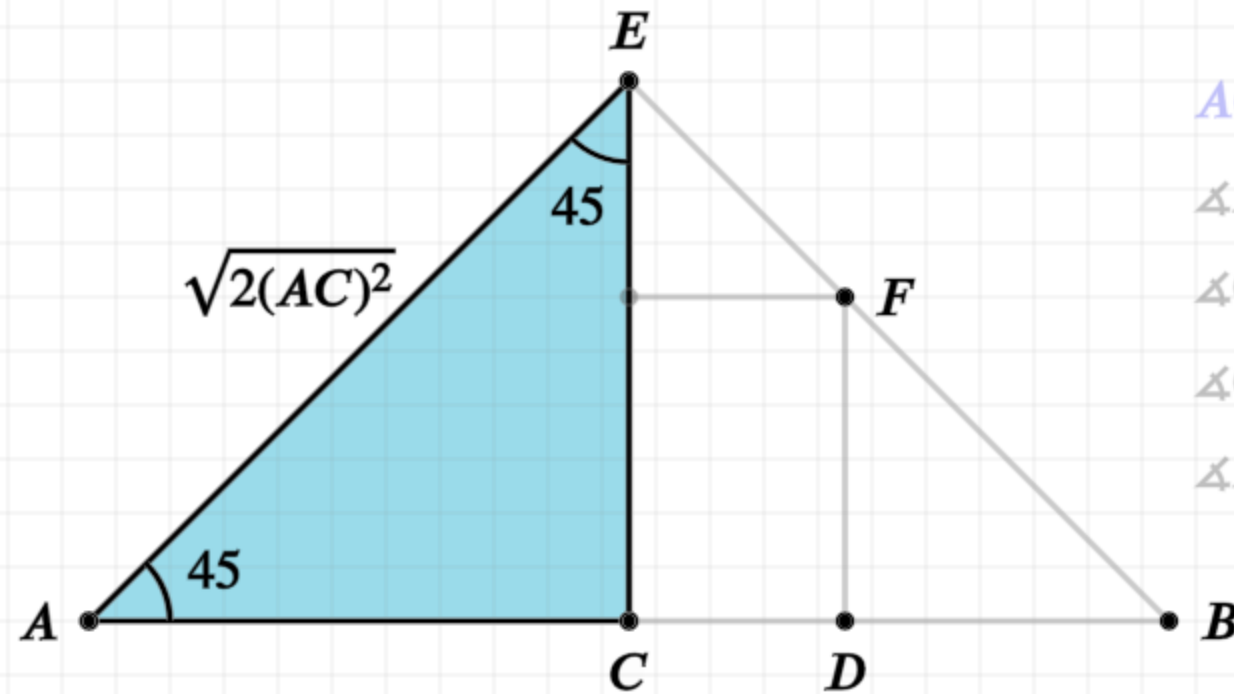
$$EG = GF$$

$$DB = FD$$

$$AE^2 = AC^2 + EC^2$$

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.



$$AE^2 = AC^2 + EC^2 = 2 \cdot AC^2$$

$$AC = BC, \quad AB = AC + CD + DB$$

$$\angle EAC = \angle CEA = 45^\circ$$

$$\angle CEB = \angle CBE = 45^\circ, \quad \angle AEB = 90^\circ$$

$$\angle GEF = \angle EFG = 45^\circ$$

$$\angle DFB = \angle FBD = 45^\circ$$

$$AC = CE$$

$$CB = CE$$

$$EG = GF$$

$$DB = FD$$

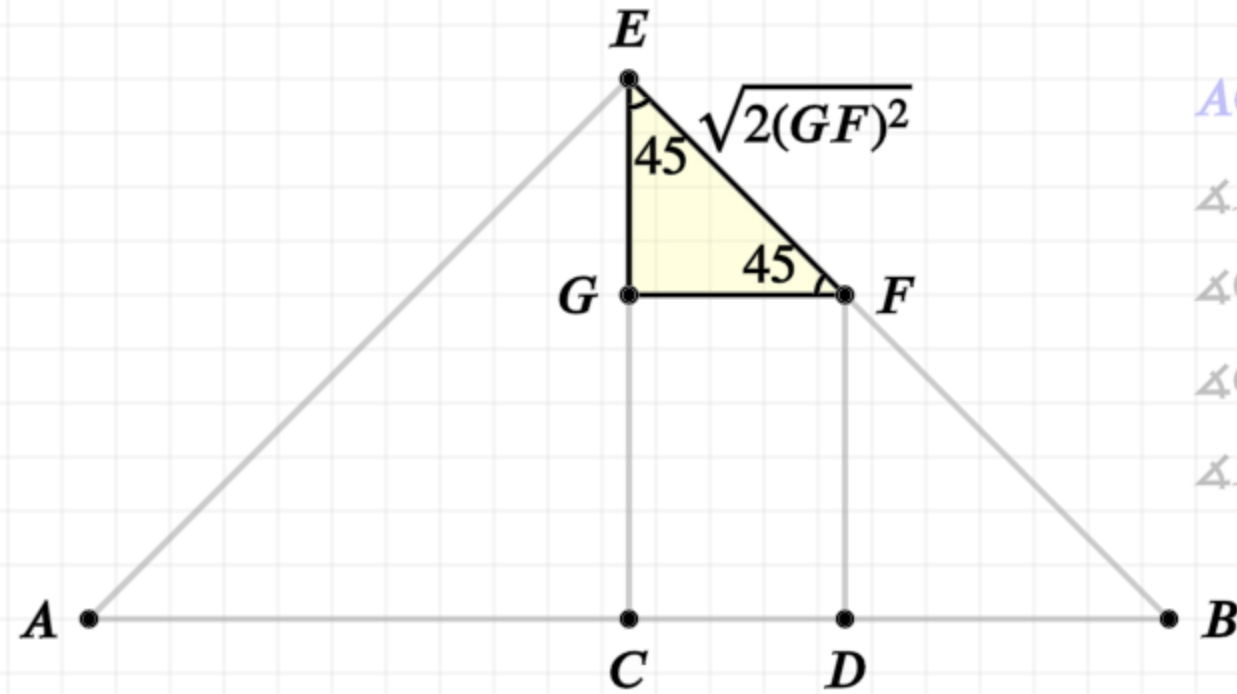
Proof (squares)

The triangle AEC is a right angle triangle, thus the square on AE equals the sum of the squares of AC and CE

Since AC equals CE, the sum of the squares of AC and CE equals twice the square of AC

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.



$$AE^2 = AC^2 + EC^2 = 2 \cdot AC^2$$

$$EF^2 = EG^2 + GF^2 = 2 \cdot GF^2$$

$$AC = BC, \quad AB = AC + CD + DB$$

$$\angle EAC = \angle CEA = 45^\circ$$

$$\angle CEB = \angle CBE = 45^\circ, \quad \angle AEB = \angle$$

$$\angle GEF = \angle EFG = 45^\circ$$

$$\angle DFB = \angle FBD = 45^\circ$$

$$AC = CE$$

$$CB = CE$$

$$EG = GF$$

$$DB = FD$$

Proof (squares)

The triangle AEC is a right angle triangle, thus the square on AE equals the sum of the squares of AC and CE

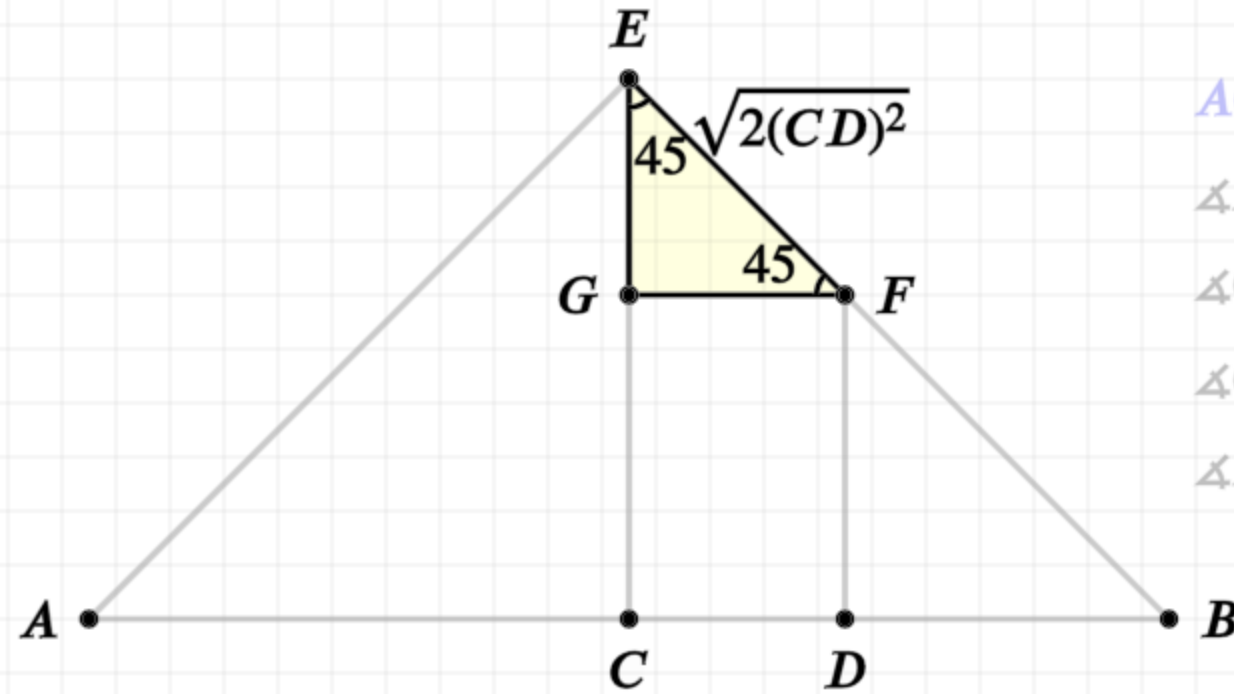
Since AC equals CE, the sum of the squares of AC and CE equals twice the square of AC

Since EG equals GF, the sum of the squares of EG and GF equals twice the square of GF

The triangle EGF is a right angle, thus the square on EF equals the sum of the squares of EG and GF

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.



$$AE^2 = AC^2 + EC^2 = 2 \cdot AC^2$$

$$EF^2 = EG^2 + GF^2 = 2 \cdot GF^2$$

$$EF^2 = 2 \cdot CD^2$$

$$AC = BC, \quad AB = AC + CD + DB$$

$$\angle EAC = \angle CEA = 45^\circ$$

$$\angle CEB = \angle CBE = 45^\circ, \quad \angle AEB = 90^\circ$$

$$\angle GEF = \angle EFG = 45^\circ$$

$$\angle DFB = \angle FBD = 45^\circ$$

$$AC = CE$$

$$CB = CE$$

$$EG = GF$$

$$DB = FD$$

Proof (squares)

The triangle AEC is a right angle triangle, thus the square on AE equals the sum of the squares of AC and CE

Since AC equals CE, the sum of the squares of AC and CE equals twice the square of AC

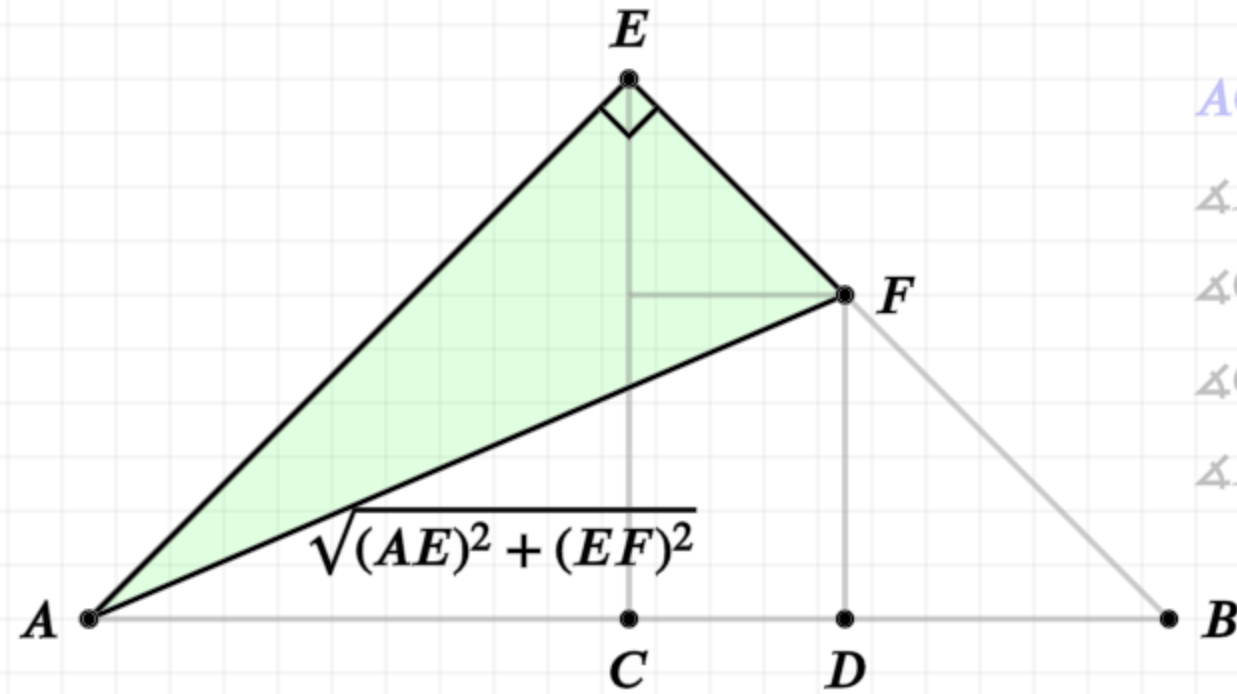
Since EG equals GF, the sum of the squares of EG and GF equals twice the square of GF

The triangle EGF is a right angle, thus the square on EF equals the sum of the squares of EG and GF

GF equals CD (I.34), thus the square on EF equals twice the sum of CD

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.



$$AC = BC, \quad AB = AC + CD + DB$$

$$\angle EAC = \angle CEA = 45^\circ$$

$$\angle CEB = \angle CBE = 45^\circ, \quad \angle AEB = 90^\circ$$

$$\angle GEF = \angle EFG = 45^\circ$$

$$\angle DFB = \angle FBD = 45^\circ$$

$$AC = CE$$

$$CB = CE$$

$$EG = GF$$

$$DB = FD$$

$$AE^2 = AC^2 + EC^2 = 2 \cdot AC^2$$

$$EF^2 = EG^2 + GF^2 = 2 \cdot GF^2$$

$$EF^2 = 2 \cdot CD^2$$

$$AF^2 = AE^2 + EF^2$$

Proof (squares)

The triangle AEC is a right angle triangle, thus the square on AE equals the sum of the squares of AC and CE

Since AC equals CE, the sum of the squares of AC and CE equals twice the square of AC

Since EG equals GF, the sum of the squares of EG and GF equals twice the square of GF

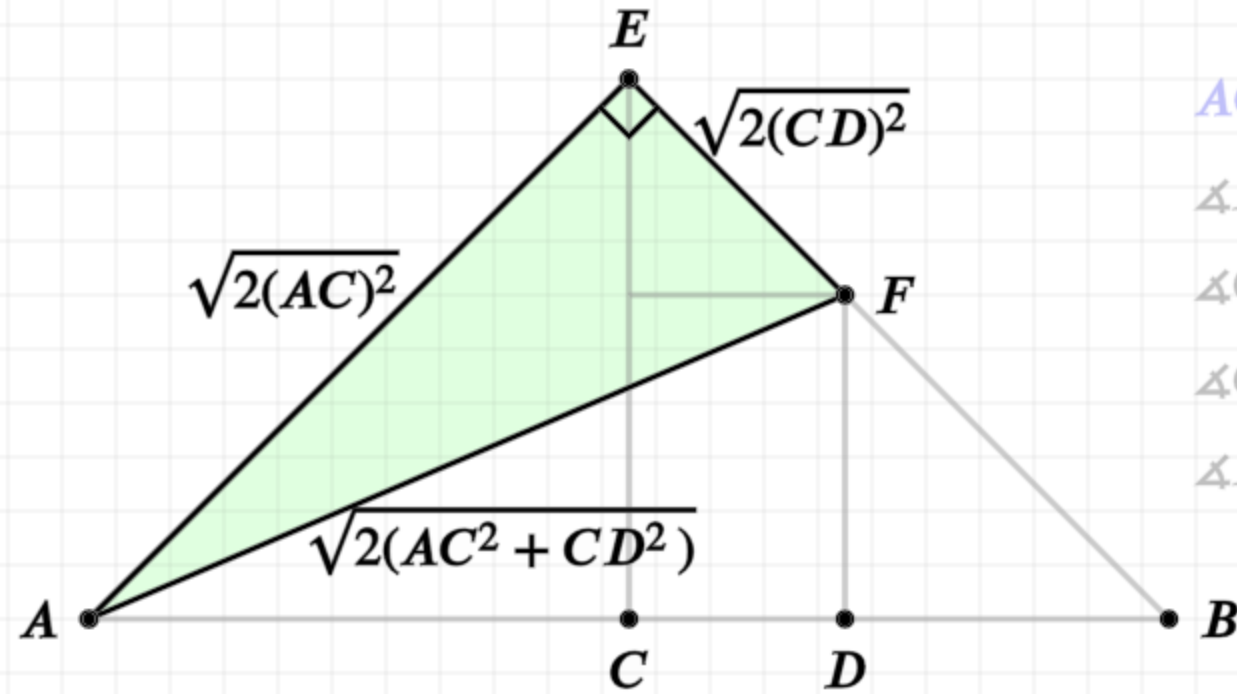
The triangle EGF is a right angle, thus the square on EF equals the sum of the squares of EG and GF

GF equals CD (I.34), thus the square on EF equals twice the sum of CD

The triangle EAF is a right angle, thus the square on AF equals the sum of the squares of AE and EF

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.



$$AE^2 = AC^2 + EC^2 = 2 \cdot AC^2$$

$$EF^2 = EG^2 + GF^2 = 2 \cdot GF^2$$

$$EF^2 = 2 \cdot CD^2$$

$$AF^2 = AE^2 + EF^2$$

$$AF^2 = 2 \cdot AC^2 + 2 \cdot CD^2 = 2(AC^2 + CD^2)$$

$$AC = BC, \quad AB = AC + CD + DB$$

$$\angle EAC = \angle CEA = 45^\circ$$

$$\angle CEB = \angle CBE = 45^\circ, \quad \angle AEB = \angle$$

$$\angle GEF = \angle EFG = 45^\circ$$

$$\angle DFB = \angle FBD = 45^\circ$$

$$AC = CE$$

$$CB = CE$$

$$EG = GF$$

$$DB = FD$$

Proof (squares)

The triangle AEC is a right angle triangle, thus the square on AE equals the sum of the squares of AC and CE

Since AC equals CE, the sum of the squares of AC and CE equals twice the square of AC

Since EG equals GF, the sum of the squares of EG and GF equals twice the square of GF

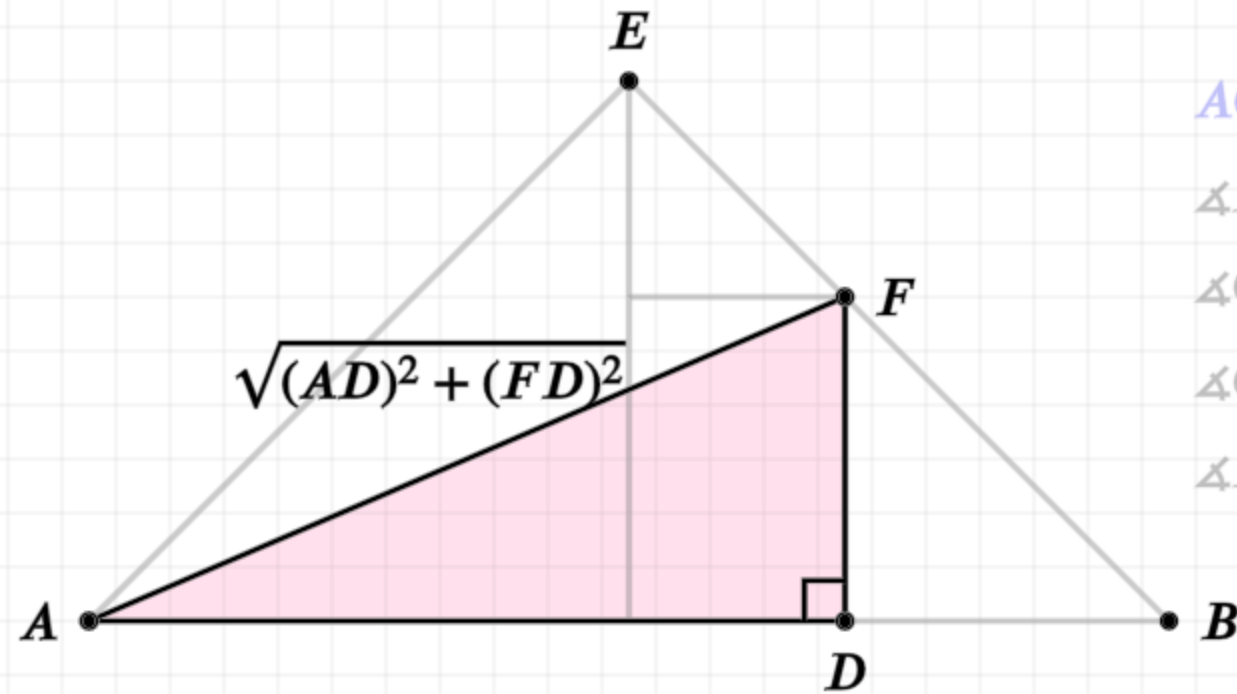
The triangle EGF is a right angle, thus the square on EF equals the sum of the squares of EG and GF

GF equals CD (I.34), thus the square on EF equals twice the sum of CD

The triangle EAF is a right angle, thus the square on AF equals the sum of the squares of AE and EF

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.



$$AC = BC, \quad AB = AC + CD + DB$$

$$\angle EAC = \angle CEA = 45^\circ$$

$$\angle CEB = \angle CBE = 45^\circ, \quad \angle AEB = 90^\circ$$

$$\angle GEF = \angle EFG = 45^\circ$$

$$\angle DFB = \angle FBD = 45^\circ$$

$$AC = CE$$

$$CB = CE$$

$$EG = GF$$

$$DB = FD$$

$$AE^2 = AC^2 + EC^2 = 2 \cdot AC^2$$

$$EF^2 = EG^2 + GF^2 = 2 \cdot GF^2$$

$$EF^2 = 2 \cdot CD^2$$

$$AF^2 = AE^2 + EF^2$$

$$AF^2 = 2 \cdot AC^2 + 2 \cdot CD^2 = 2(AC^2 + CD^2)$$

$$AF^2 = AD^2 + FD^2$$

Proof (squares)

The triangle AEC is a right angle triangle, thus the square on AE equals the sum of the squares of AC and CE

Since AC equals CE, the sum of the squares of AC and CE equals twice the square of AC

Since EG equals GF, the sum of the squares of EG and GF equals twice the square of GF

The triangle EGF is a right angle, thus the square on EF equals the sum of the squares of EG and GF

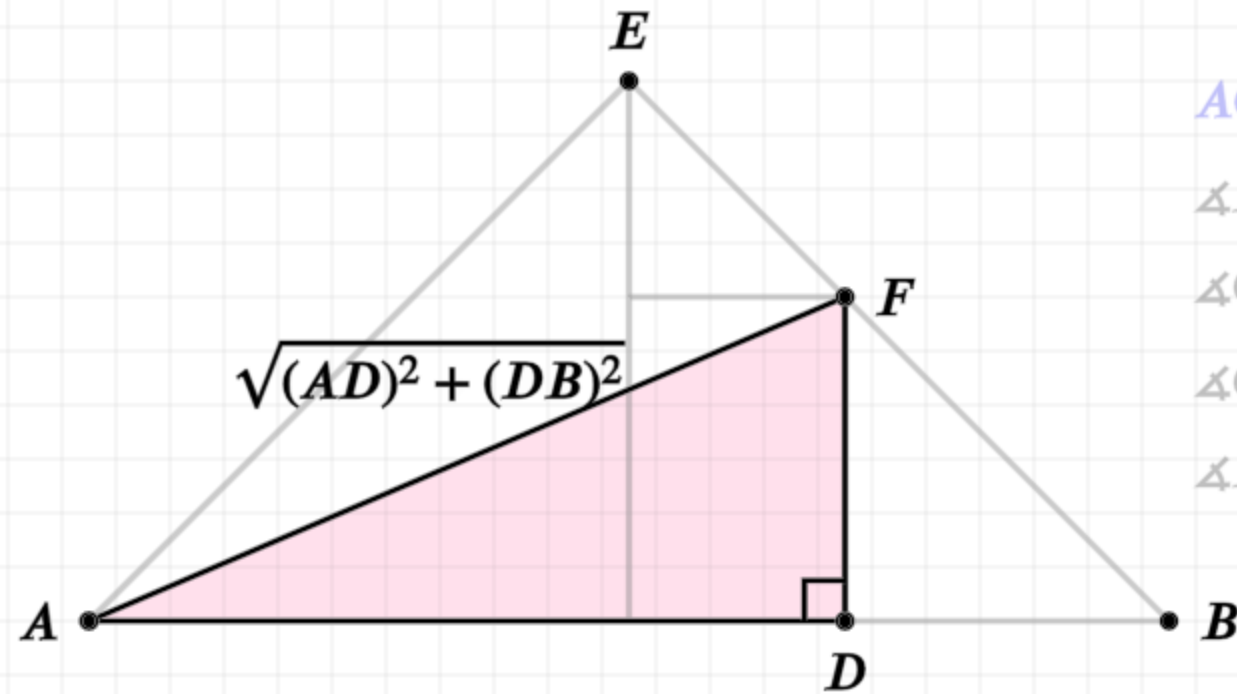
GF equals CD (I.34), thus the square on EF equals twice the sum of CD

The triangle EAF is a right angle, thus the square on AF equals the sum of the squares of AE and EF

The triangle FAD is a right angle triangle, thus the square on AF equals the sum of the squares of AD and DF

Proposition 9 of Book 2

If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.



$$AC = BC, \quad AB = AC + CD + DB$$

$$\angle EAC = \angle CEA = 45^\circ$$

$$\angle CEB = \angle CBE = 45^\circ, \quad \angle AEB = 90^\circ$$

$$\angle GEF = \angle EFG = 45^\circ$$

$$\angle DFB = \angle FBD = 45^\circ$$

$$AC = CE$$

$$CB = CE$$

$$EG = GF$$

$$DB = FD$$

$$AE^2 = AC^2 + EC^2 = 2 \cdot AC^2$$

$$EF^2 = EG^2 + GF^2 = 2 \cdot GF^2$$

$$EF^2 = 2 \cdot CD^2$$

$$AF^2 = AE^2 + EF^2$$

$$AF^2 = 2 \cdot AC^2 + 2 \cdot CD^2 = 2(AC^2 + CD^2)$$

$$AF^2 = AD^2 + FD^2$$

$$AF^2 = AD^2 + DB^2$$

Proof (squares)

The triangle AEC is a right angle triangle, thus the square on AE equals the sum of the squares of AC and CE

Since AC equals CE, the sum of the squares of AC and CE equals twice the square of AC

Since EG equals GF, the sum of the squares of EG and GF equals twice the square of GF

The triangle EGF is a right angle, thus the square on EF equals the sum of the squares of EG and GF

GF equals CD (I.34), thus the square on EF equals twice the sum of CD

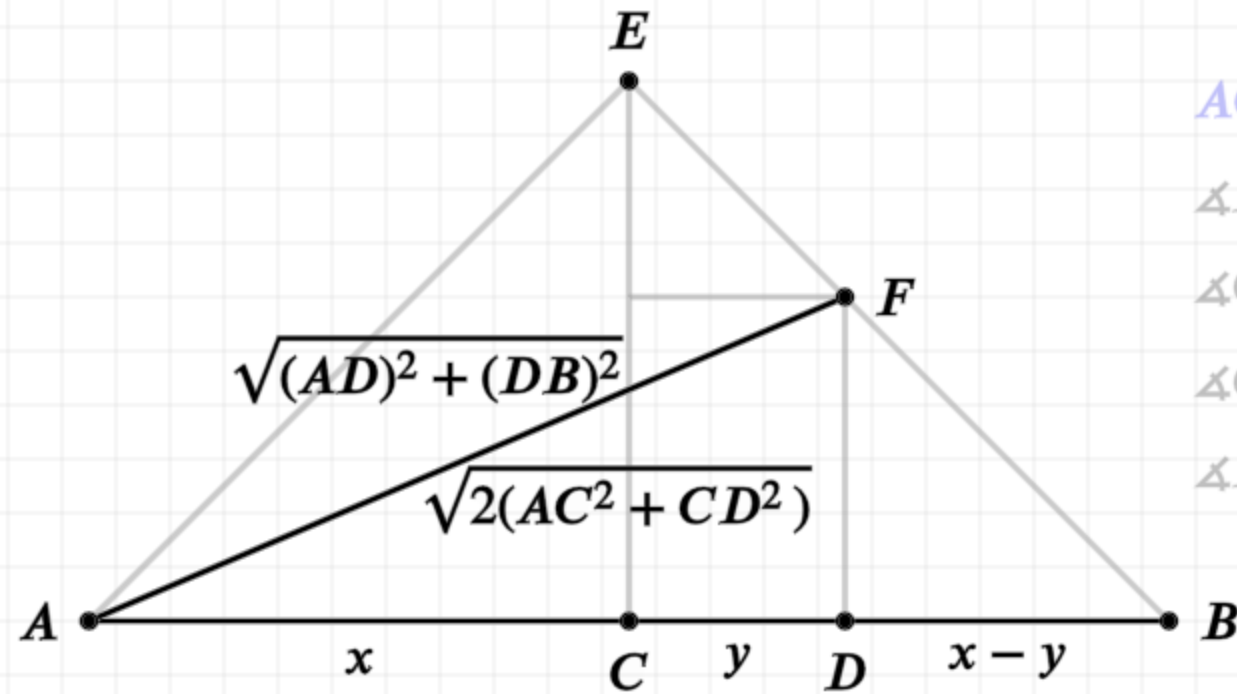
The triangle EAF is a right angle, thus the square on AF equals the sum of the squares of AE and EF

The triangle FAD is a right angle triangle, thus the square on AF equals the sum of the squares of AD and DF

But DF equals DB, so the square of AF is the sum of the squares of AD and BD

Proposition 9 of Book 2

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$$AE^2 = AC^2 + EC^2 = 2 \cdot AC^2$$

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$$AF^2 = AD^2 + FD^2$$

$$AF^2 = AD^2 + DB^2$$

$$AD^2 + DB^2 = 2(AC^2 + CD^2) \rightarrow (x+y)^2 + (x-y)^2 = 2(x^2 + y^2)$$

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$$\angle CEB = \angle CBE = 45^\circ, \quad \angle AEB = \angle$$

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Proof (squares)

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The triangle EAF is a right angle, thus the square on AF equals the sum of the squares of AE and EF

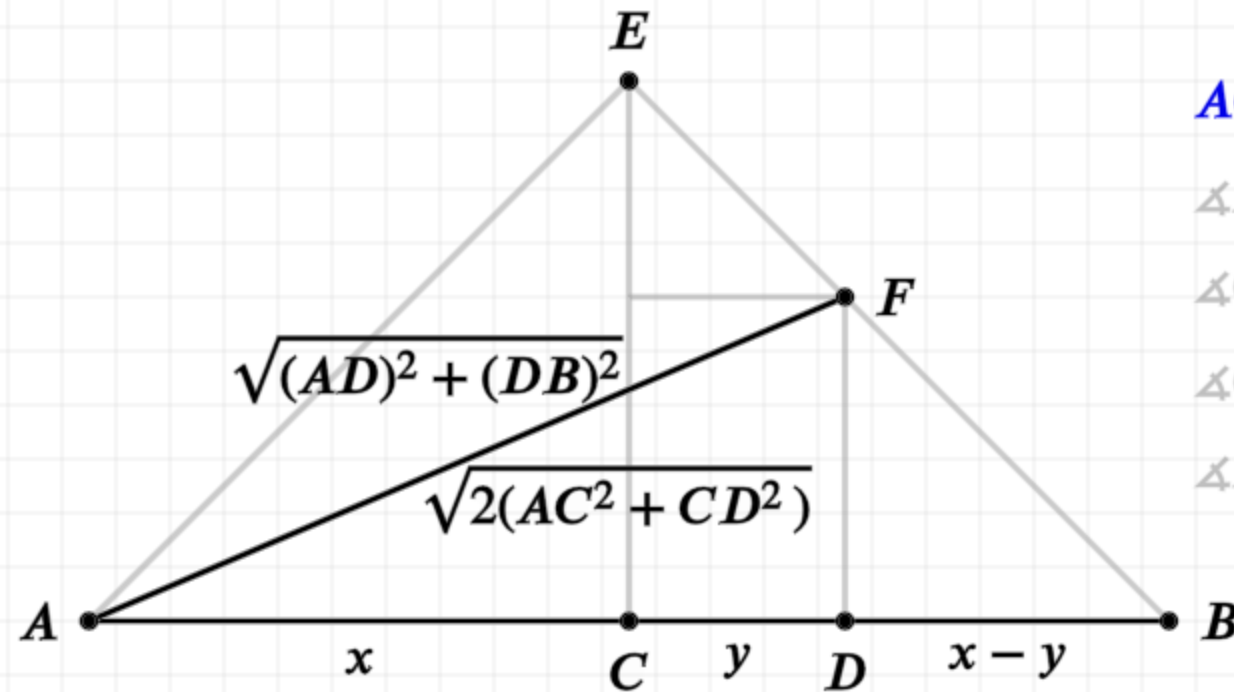
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Rearranging the equalities gives the original proposition

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$$AF^2 = AD^2 + DB^2$$

$$AD^2 + DB^2 = 2(AC^2 + CD^2) \rightarrow (x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$$

Proof (squares)

The triangle AEC is a right angle triangle, thus the square on AE equals the sum of the squares of AC and CE

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Rearranging the equalities gives the original proposition

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